Ablation studies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Remove one component. Preserve the controls. Inspect the failures.

Explain conditional ablation differences, missing interactions and critical failure categories.

Keep one habit: Comparison contract → main condition ledger → native missing-cell grid → separate filter trace → interaction counterexamples → independent fifteen-case task.

MAIN LESSON · SECTION 1

Remove one component and inspect what changes

> A small score change can conceal a critical failure category.

Learning goals

I can explain the **mechanism** and terminology of **ablation** studies.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve controlled comparisons and practical-effect limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting response

- Recall Multiple comparisons. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **ablation** studies using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **ablation** studies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define what is removed and what must stay fixed

> Separate total performance, failure severity and interaction evidence.

Begin with a component question

To investigate a component, compare the complete system with a version that removes only that component. Use the same evaluation cases, scoring and remaining settings. The difference describes this removal in this configuration.

It does not assign a permanent score to the component: retrieval may be useful only when another component uses its results, and a validation step may protect an important boundary even when the total score barely changes.

Original ablation definition

Ablation removes a component to investigate its contribution under a controlled comparison. **Interaction**s can make contributions depend on other components.

Complete controlled-comparison contract

Comparison item	Complete constructed contract
Common evaluation	Same ten cases, same scoring and all other settings fixed

Comparison item	Complete constructed contract
Full system	Retrieval and validation present; 8/10 passes
Remove retrieval only	Validation remains; 5/10 passes
Remove validation only	Retrieval remains; 7/10 passes; an unauthorized action is stipulated
Remove both	Not supplied; do not invent this outcome
Reported effects	Full minus ablated pass count and percentage-point difference
Case identities	Not supplied; net differences do not identify which cases flipped
Scope	Supplied fictional summaries, not actual model runs or population causal estimates

Given and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace both removals without adding their effects

> The missing joint-removal condition matters.

Original comparison procedure

Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.

All supplied conditions and remaining unknowns

Condition	Pass count/rate	Net drop from full	Interpretation
Full system	8/10 = 80%	Reference	Both components present
Without retrieval	5/10 = 50%	3 passes; 30 percentage points	Conditional removal effect in this configuration
Without validation	7/10 = 70%	1 pass; 10 percentage points	Also permits a stipulated unauthorized action
Without both	Unknown	Unknown	Cannot calculate joint effect or interaction
Changed case identities	Unknown	Net count is not a flip ledger	Inspect per-case records if collected

State controlled settings and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Ablation studies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action.	Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.	Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict why removing a filter and changing the model confounds attribution

> The changed condition varies two factors at once.

Main task and response

- Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed picture condition

- Change: Remove the filter and also change the model version.

▼ **Check after predicting**

Prediction: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the ablation grid and the separate filter example

> Do not mix the main validation boundary with the separate filter counts.

Main supplied counts and checked result

Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action.

Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate filter facts and scope

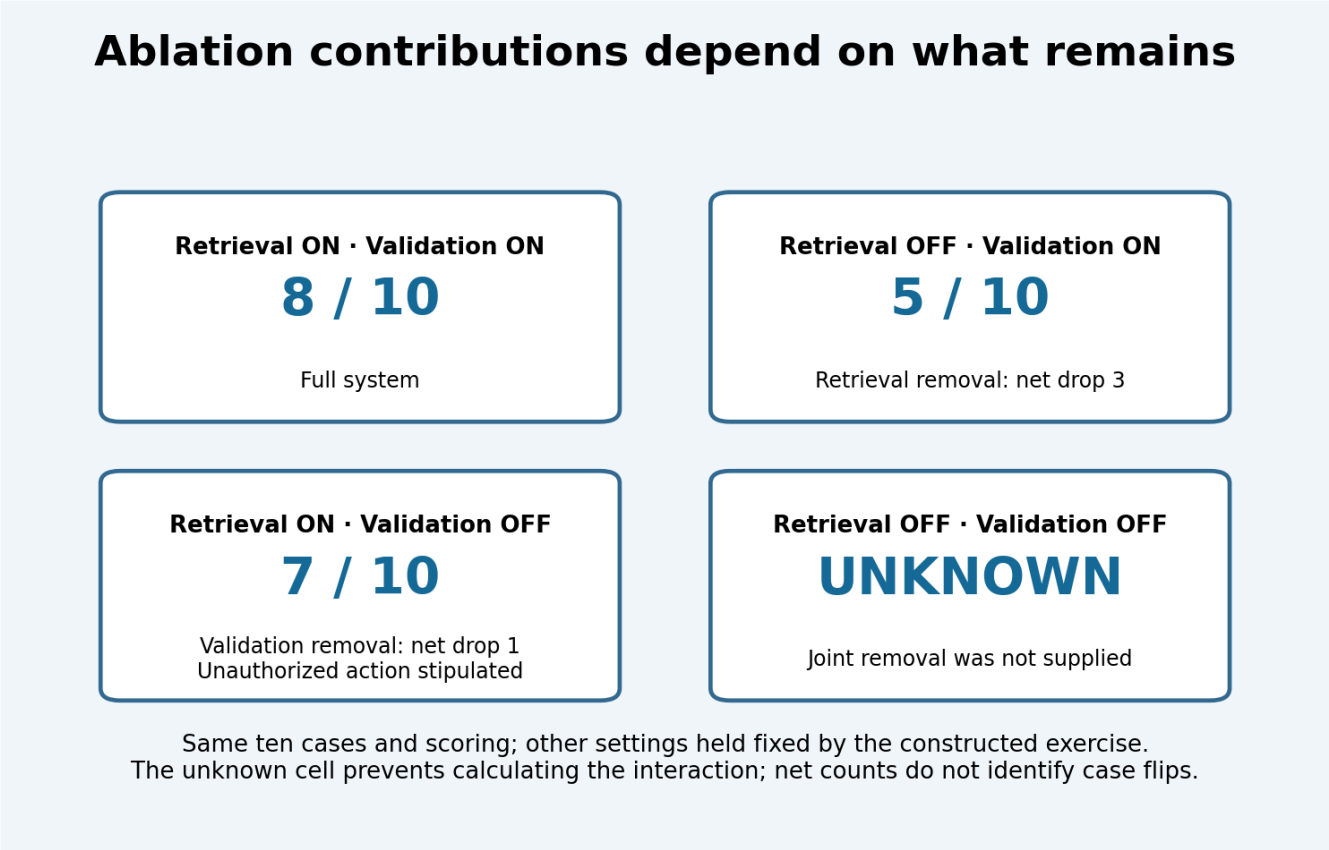
Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

The main grid separates retrieval and validation. Its two observed net score drops are three and one passes, but their sum does not establish the result of removing both. The separate picture investigates a relevance filter instead: full pipeline 8/10 versus filter removed 6/10, a net drop of two passes or twenty percentage points.

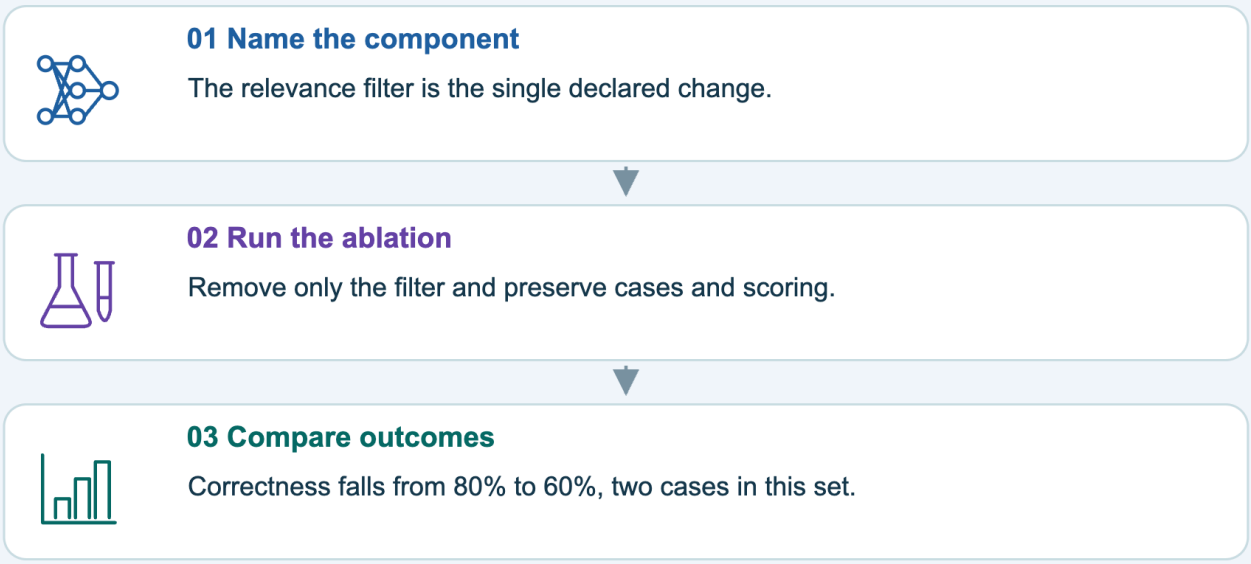
The aggregate counts do not show whether exactly two cases changed; gains and losses could offset. If the model version also changes, the comparison cannot isolate the filter. Neither picture records a real model execution or proves a universal contribution.

See the missing joint-removal condition



AI ACADEMY / AI-7/16 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the component	The relevance filter is the single declared change.
2. Run the ablation	Remove only the filter and preserve cases and scoring.
3. Compare outcomes	Correctness falls from 80% to 60%, two cases in this set.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Complete filter arithmetic and interpretation

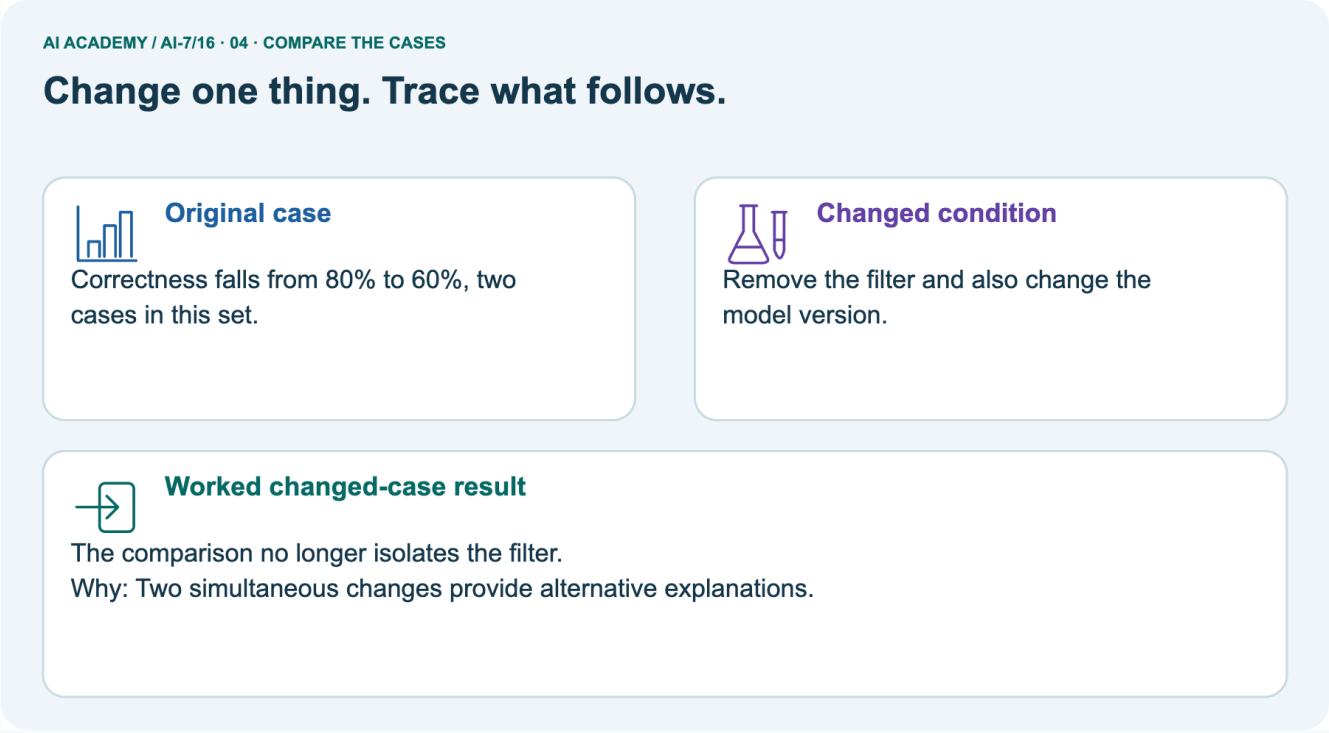
Separate filter picture	Calculation or evidence limit
Full pipeline	8/10 = 80%
Filter removed only	6/10 = 60%
Net difference	8 – 6 = 2 passes; 80% – 60% = 20 percentage points
Relative decrease from full	2/8 = 25%; differs from 20 percentage points
Changed model as well	Two factors vary; filter-only attribution is lost

Separate filter picture	Calculation or evidence limit
Which cases changed	Not supplied; net two is not proof of exactly two flips
General conclusion	Conditional descriptive comparison; no universal benefit or significance established

Predict changing the model too

- Change one thing: Remove the filter and also change the model version. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining limits

- **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

- Read the labelled picture: Calculate the filter **ablation**'s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

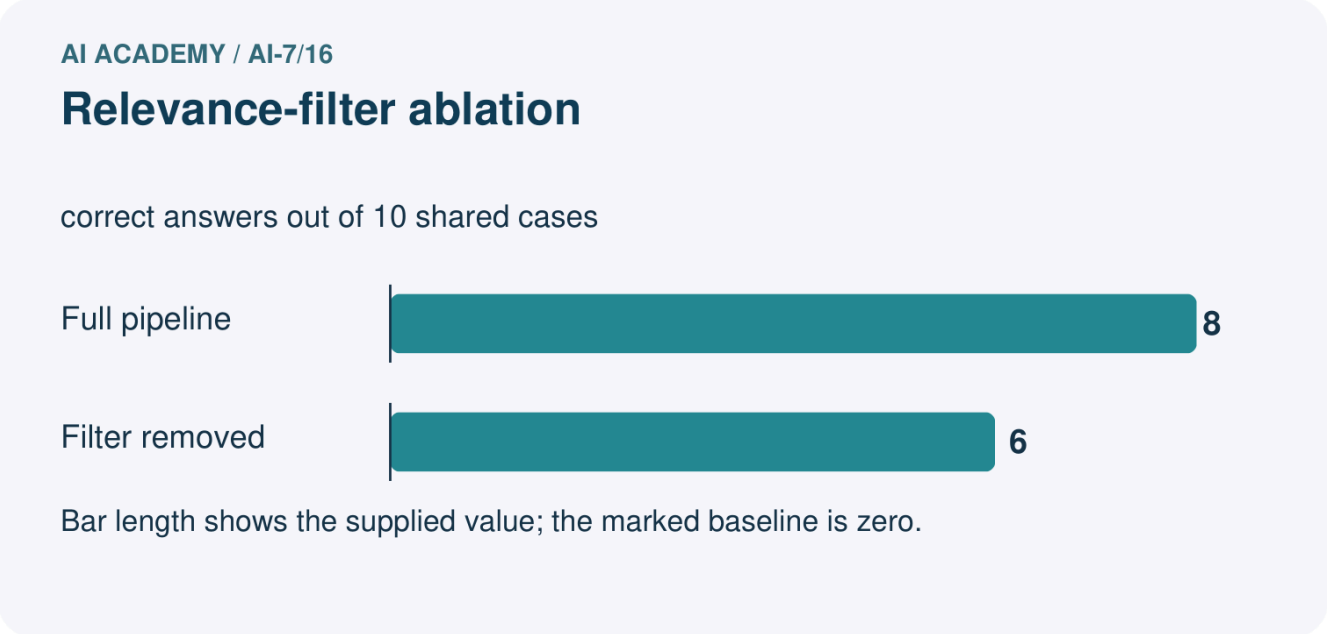
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Ablation studies

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/16

Ablation studies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the component

The relevance filter is the single declared change.



2. Run the ablation

Remove only the filter and preserve cases and scoring.



3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

CHANGE ONE THING

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair additive contributions and score-only judgments

> Interactions and critical failures can matter more than a small aggregate change.

Original limitation and repair setup

Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Missing-cell and severity probes

Ablation audit	Checked interpretation
Removing validation drops only one pass	Do not dismiss the stipulated unauthorized action
Adding the two observed drops	Does not determine the unobserved joint removal
Both-removal count hypothetically 4	Interaction contrast 8 − 7 − 5 + 4 = 0
Both-removal count hypothetically 6	Interaction contrast 8 − 7 − 5 + 6 = 2; also compatible with supplied three cells
Hypothetical counterexamples	Neither is an observed result; they demonstrate missing information
Net score falls by two	Per-case gains and losses may offset; exact flips unknown
Filter and model both change	Alternative explanations prevent filter-only attribution
All settings said to be fixed	An exercise assumption; actual study would need recorded settings and outputs

Write a supported repair

- A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise tracing removal effects and naming missing evidence

> Reserve the fifteen-case comparison for the first independent attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example "**Ablation** studies". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual check

- Use the worked case for **Ablation** studies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the filter **ablation**’s two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

> **My explanation using a pictured step and the given facts**

Reserve the selected new case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run a descriptive ablation audit without inventing missing runs

> Use counterexamples to explain unknown interactions.

Current investigation

Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

Complete executable arithmetic and counterexample

```
"""Descriptive ablation arithmetic on supplied summaries; no real AI actions."""
from fractions import Fraction

def ablation_drop(full, removed, total):
 if any(type(x) is not int for x in (full,removed,total)) or total<=0 or not
0<=full<=total or not 0<=removed<=total:
  raise ValueError('require integer counts within a positive common denominator')
 return {'count_drop':full-removed,'percentage_point_drop':100*Fraction(full-
removed,total)}
main_retrieval=ablation_drop(8,5,10)
main_validation=ablation_drop(8,7,10)
picture=ablation_drop(8,6,10)
assert main_retrieval['count_drop']==3 and main_validation['count_drop']==1
assert picture['percentage_point_drop']==20
# Two possible joint-removal outcomes, neither supplied by the actual exercise.
# Full R+V = 8; R only = 7; V only = 5. Neither component = unknown.
def interaction_contrast(neither):
 if type(neither) is not int or not 0<=neither<=10:raise ValueError('invalid hypothetical
count')
 return 8-7-5+neither
assert interaction_contrast(4)==0 and interaction_contrast(6)==2
print('Main retrieval removal:',main_retrieval)
print('Main validation removal:',main_validation)
print('Separate filter picture:',picture)
print('Main no-validation condition also permits a stipulated unauthorized action; a one-
pass drop does not make that boundary unimportant.')
print('Joint-removal outcome is UNKNOWN. Hypothetical neither counts 4 versus 6 give
interaction contrasts 0 versus 2; these are counterexamples, not observations.')
print('Counts give net changes, not the identities of changed cases. No model, study,
external action, significance test or universal causal attribution is performed.')
```

Predictions, results and limits

1. Predict the two main pass drops and percentage-point changes. Then inspect the complete descriptive audit. It computes supplied summary differences and shows two explicitly hypothetical joint-removal outcomes consistent with the three known cells. Their different interaction contrasts demonstrate why the missing cell cannot be inferred. The program does not simulate an unauthorized action or run a model.

   Keep hypothetical values separate from observations, and preserve the independent fifteen-case totals for a first attempt.

Evidence record

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare three conditions on fifteen cases

> Compute both net drops and state why contributions are not universally additive.

Independent case

Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Full 12/15, without A 9/15, without B 11/15. Compute sample drops and state **interaction** limit.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save each condition, failure category and unchanged setting

> Include missing conditions rather than silently assuming their outcomes.

Project and artifact

Add a labelled **ablation** studies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **ablation** studies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local arithmetic or paper trace

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain a conditional removal effect and a missing interaction

> Counts, controls and failure severity belong in the same explanation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **ablation** studies

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Ablation	Removing or disabling a component in a comparison
Conditional contribution	An effect given the other components
Interaction	An effect depending on another factor's setting

Use two terms to explain a step in this week’s example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible **interaction**s.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Ablation studies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Full system 8/10, without retrieval 5/10, without validation 7/10; the last permits an unauthorised action.	Compare full system, without retrieval and without validation on identical cases and scoring. Hold other settings fixed; record changed failure categories, not just totals.	Drops 3 and 1 passes respectively in this sample. Validation's critical action boundary is not captured by pass-count magnitude alone.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution.

By the end, I can	How I will show it
Explain the mechanism and terminology of ablation studies	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter’s contribution.
- Which labelled step matches this detail? Correctness falls from **80%** to **60%**, two cases in this set.
- 1. Name the component
- 2. Run the **ablation**
- 3. Compare outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare outcomes Correctness falls from **80%** to **60%**, two cases in this set. An **ablation** probes a component by removing or changing it while holding the rest fixed.

It supports a scoped comparison, not a claim that the component always helps.

Changed-case question and optional hint

- Changed-case practice: Remove the filter and also change the model version. Which conclusion is supported by the supplied facts?
- 1. The comparison no longer isolates the filter.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

Repair and retain the record

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Ablation studies: the case



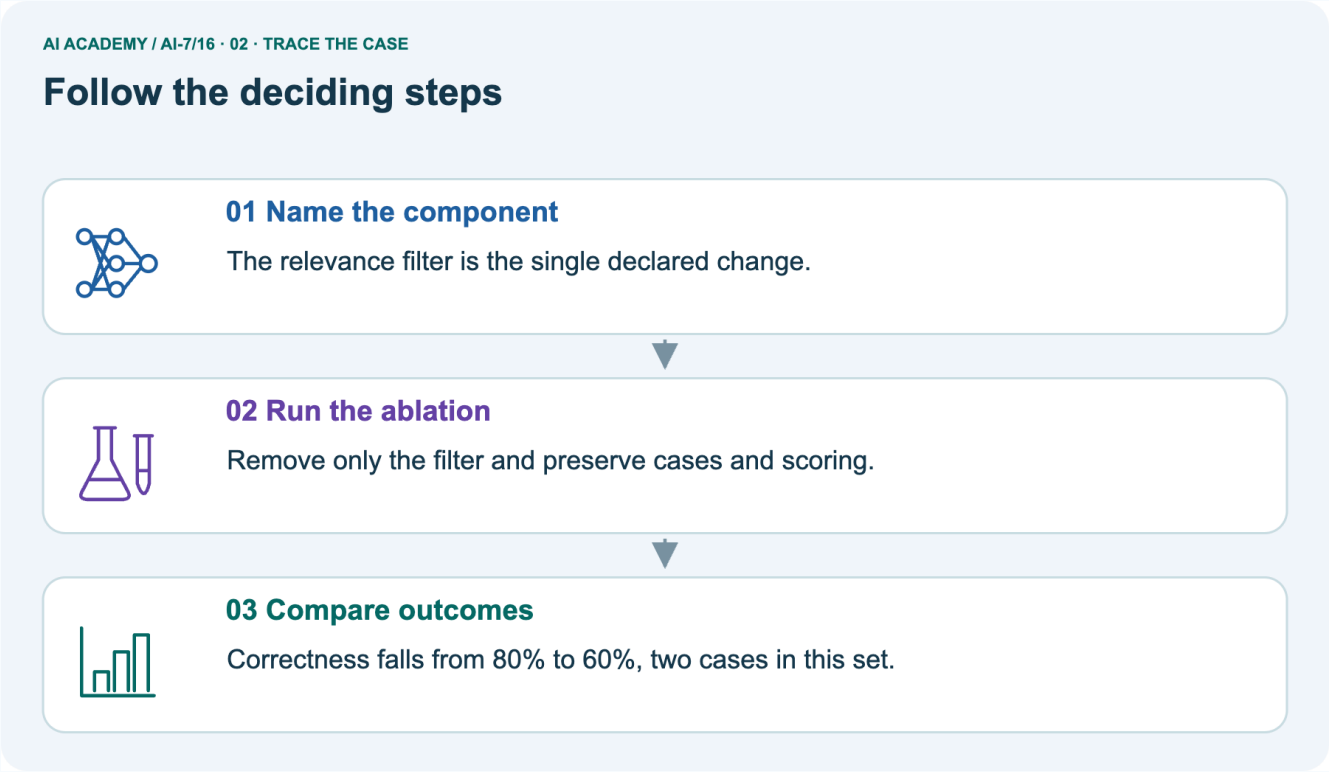
Follow the labelled picture with the matching explanation. Open the vector diagram

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution. Supplied facts: Full pipeline: 8/10 correct.

Filter removed: 6/10 on the same cases, with all other settings fixed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

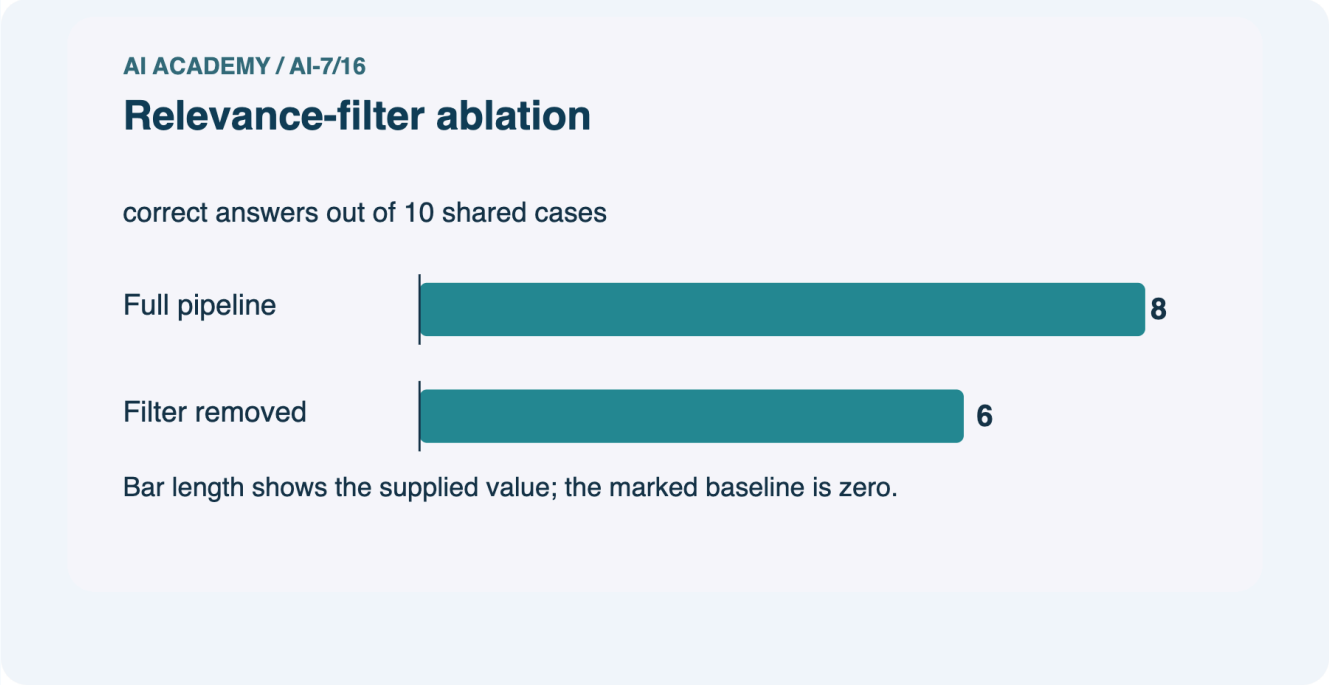


Follow the labelled picture with the matching explanation. Open the vector diagram

Name the component: The relevance filter is the single declared change. → Run the **ablation**: Remove only the filter and preserve cases and scoring. → Compare outcomes: Correctness falls from **80%** to **60%**, two cases in this set.

Point to each numbered stage and explain which recorded **input** it uses.

03. Relevance-filter ablation

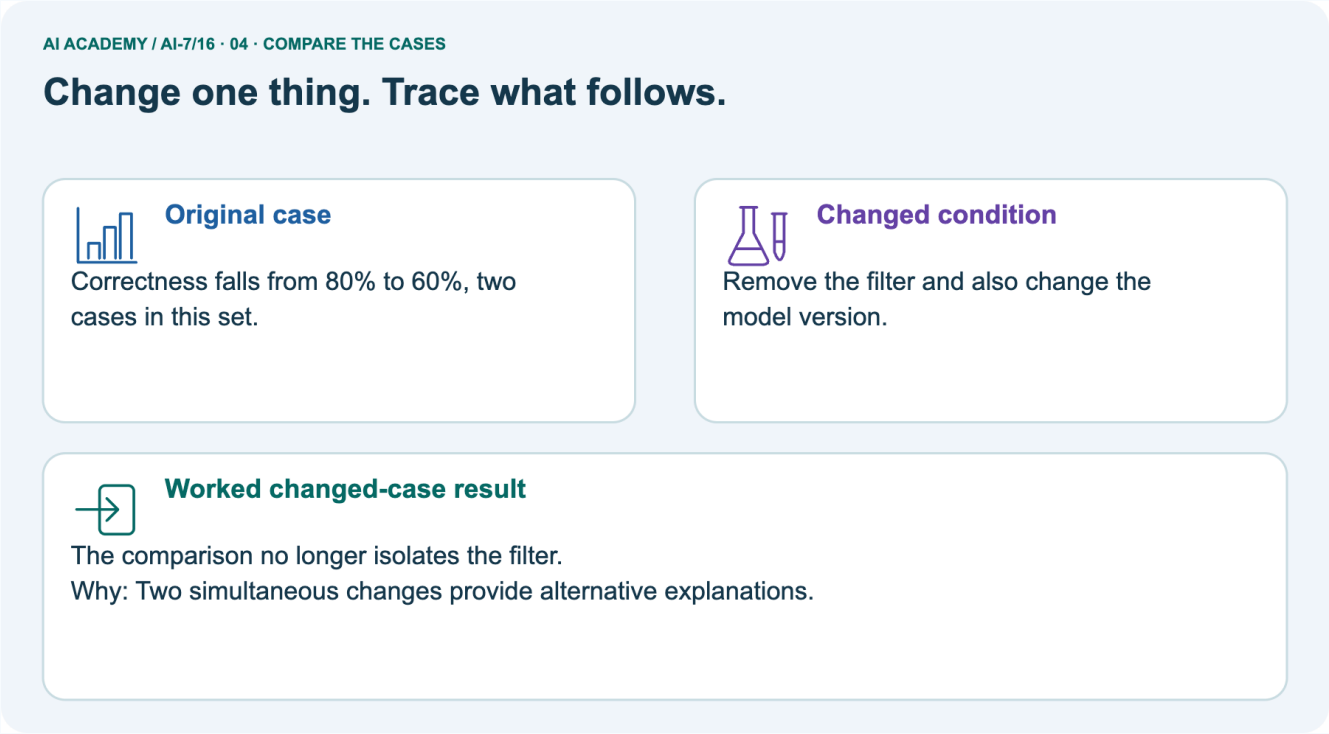


Follow the labelled picture with the matching explanation. Open the vector diagram

Relevance-filter **ablation**. Units: correct answers out of 10 shared cases. Full pipeline: 8. Filter removed: 6.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



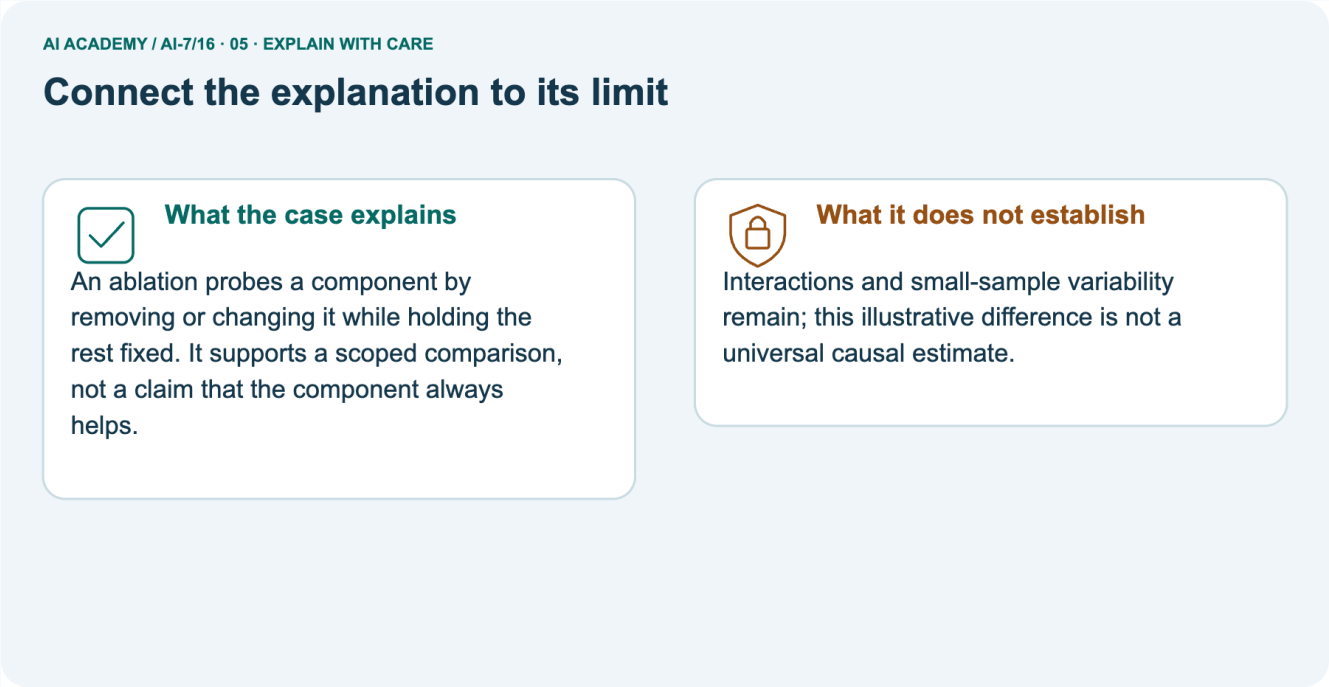
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correctness falls from **80%** to **60%**, two cases in this set. Changed condition: Remove the filter and also change the model version. Expected result: The comparison no longer isolates the filter.

Reason: Two simultaneous changes provide alternative explanations.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



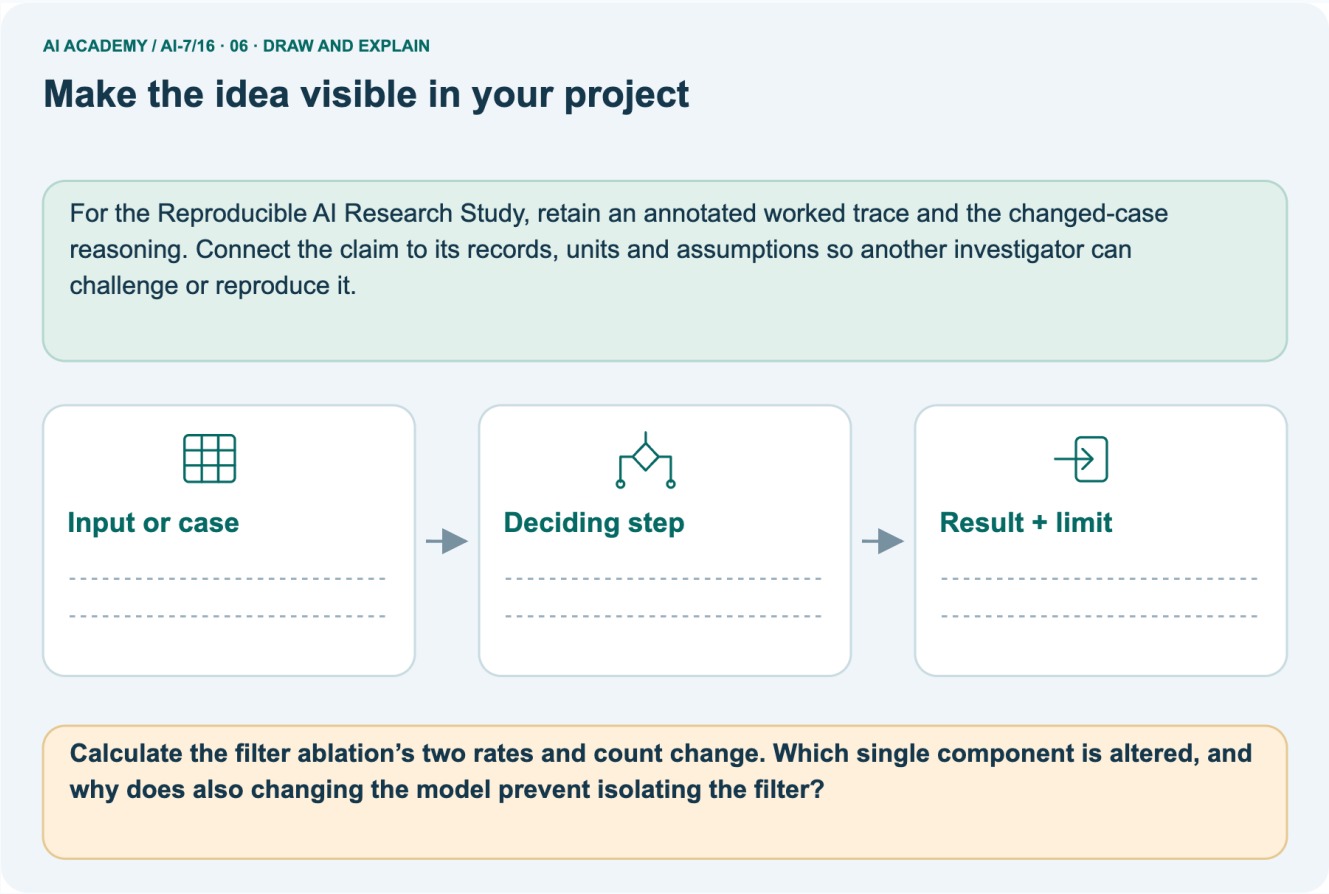
Follow the labelled picture with the matching explanation. Open the vector diagram

An **ablation** probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Limit: **Interaction**s and small-sample variability remain; this illustrative difference is not a universal causal estimate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the filter **ablation**'s two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Ablation studies: the case

02. Follow the deciding steps

03. Relevance-filter ablation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.